A stationary nilsequence with correlation identically one

Conjecture 00000001924

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Let $G = (\mathbb{R}, +)$ and $\Gamma = \mathbb{Z}$, so $G/\Gamma = \mathbb{R}/\mathbb{Z}$ is a compact one-step nilmanifold. Take the constant polynomial orbit $g(n) = 0$ and the continuous, Lipschitz observable $F \equiv 1$. Then $a_n = F(g(n)+\Gamma) = 1$ is a basic polynomial nilsequence. It is invariant under every integer shift, so it satisfies stationarity without any extra condition on the shift.

Theorem 1. *The normalized correlation of this nilsequence with the zero-frequency linear phase equals one for every $N \geq 1$ and admits no $O(N^{-1/2})$ bound.*

Proof. For $e(\alpha n) = \exp(2\pi i \alpha n)$, set $\alpha = 0$. Then

$$\frac{1}{N} \sum_{n=0}^{N-1} a_n \overline{e(0n)} = 1.$$

If the asserted decay held, some real constant C would satisfy $1 \leq CN^{-1/2}$ for all sufficiently large N. The right side tends to zero, giving $1 \leq 0$, a contradiction. □

Hypotheses. The official statement does not require a mean-zero observable, a nonresonance condition, a nonzero frequency, or subtraction of the mean. Such restrictions would change the conjecture. The witness uses the ordinary normalized correlation; starting the sum at one gives the same value.

Lean correspondence. `circleSequence` uses Mathlib's `AddCircle 1`, its actual additive rotation, and a bundled continuous observable. `witness_one` and `stationary` prove the claimed sequence and shift invariance. `correlation` uses complex exponentials, conjugation and a finite sum; `correlation_one` evaluates it for every nonzero N. `no_weyl_bound` rules out every eventual square-root bound using a proved limit.

Definition source. Green and Tao, The Möbius function is strongly orthogonal to nilsequences, and Leibman, Nilsequences, null-sequences, and multiple correlation sequences. The present problem concerns linear phases, not the Möbius function.

Verification. Lean 4.19.0 with Mathlib pinned to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The accompanying project and verification transcript give the exact theorem and axiom audit. No proof placeholders, native evaluator proof bypass, or custom axiom declarations are used. This is a prepared submission; official acceptance has not been claimed.